

The River Wave Concept: Integrating River Ecosystem Models

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We introduce the river wave concept: a simple, holistic model that unifies river ecosystem concepts. The river wave concept proposes that river flow can be conceptualized as a series of waves varying in shape, amplitude, wavelength, and frequency, traveling longitudinally and laterally; the position on the wave determines the source of organic production or inputs and the storage, transformation, and transport of material and energy; and existing concepts explain ecosystem phenomena at different positions on the river wave. The river wave concept hypothesizes that, at the troughs of waves, local autochthonous and allochthonous inputs predominate; on the ascending or descending limbs of waves, upstream allochthonous inputs and longitudinal transport of material and energy predominate; and as waves rise to crests, allochthonous inputs of material and energy and autochthonous production from the floodplain increase. We describe how river waves interact with their environment and the relevance for biota.

Keywords: river ecosystem models, riverine biota, autochthonous and allochthonous inputs, geomorphology, hydrology

Rivers, in their natural state, are among the most dynamic, diverse, and complex ecosystems on the planet. They are also probably the most degraded of all ecosystems, and there is little evidence that this will change in the near future (Dudgeon 2010). Because they are critical for human well-being, most human societies rank river conservation and management very highly. Unlike other ecosystems, however, rivers are dynamic networks of channels and floodplains, connected and disconnected through the action of flow. As a result, the patterns that exist and the processes that operate in rivers are unique and so require unique predictive models if we are to effectively conserve and manage them both for intrinsic and extrinsic reasons. It is for these reasons that river scientists have sought to understand and characterize the patterns and processes in river ecosystem functioning. Early in the twentieth century, there were attempts to conceptualize how rivers changed biologically from source to mouth, and they typically involved the differentiation of zones based on faunal attributes (e.g., fish zones; figure 1; Gerking 1945). These river zones suffered from regional idiosyncrasies, whereas generality was the goal. Following Hynes's (1970) *The Ecology of Running Waters*, several major concepts and their spinoffs have emerged. These concepts are intended to holistically describe the sources of energy (autochthonous or allochthonous), how production ratios (autotrophic: heterotrophic and photosynthesis: respiration) change longitudinally, laterally, and with discharge (Vannote et al. 1980, Junk et al. 1989, Thorp and DeLong 1994, 2002, Thorp et al. 2006, 2008). The

development of river ecosystem function ideas has been reviewed by others, and we direct the reader to these sources (Thorp et al. 2006, 2008, Winemiller et al. 2010). Here, we briefly revisit the main concepts and their underlying principles; discuss attempts at synthesis; and present a case for a new, simple, unifying model.

The river continuum concept (Vannote et al. 1980) was the first model created to conceptualize the sources and transport of carbon and energy in river ecosystems. The river continuum concept emphasized the longitudinal links in a river, combining stochastic, abiotic (geomorphology and hydrology) and deterministic, biotic (trophic relationships) aspects and primarily involving upstream inputs of organic matter and its processing by macroinvertebrates. The flood pulse concept came soon after and arose from a dissatisfaction with the generality of the river continuum concept and its focus on the patterns and processes along the longitudinal axis of permanent, lotic riverine environments and was based on observations largely derived from tropical floodplain rivers (Junk et al. 1989), although it was extended to temperate systems (Tockner et al. 2000) and to lakes (Wantzen et al. 2008). The flood pulse concept emphasizes the floodplain as the primary source of material and energy that fuels food webs in floodplain rivers. Inundation of the floodplain by a flood pulse is the catalyst for material transport and primary production and for movement of that material and energy from the floodplain into the main channel. The emphasis, therefore, is more on lateral connectivity than on a longitudinal continuum, and the flood

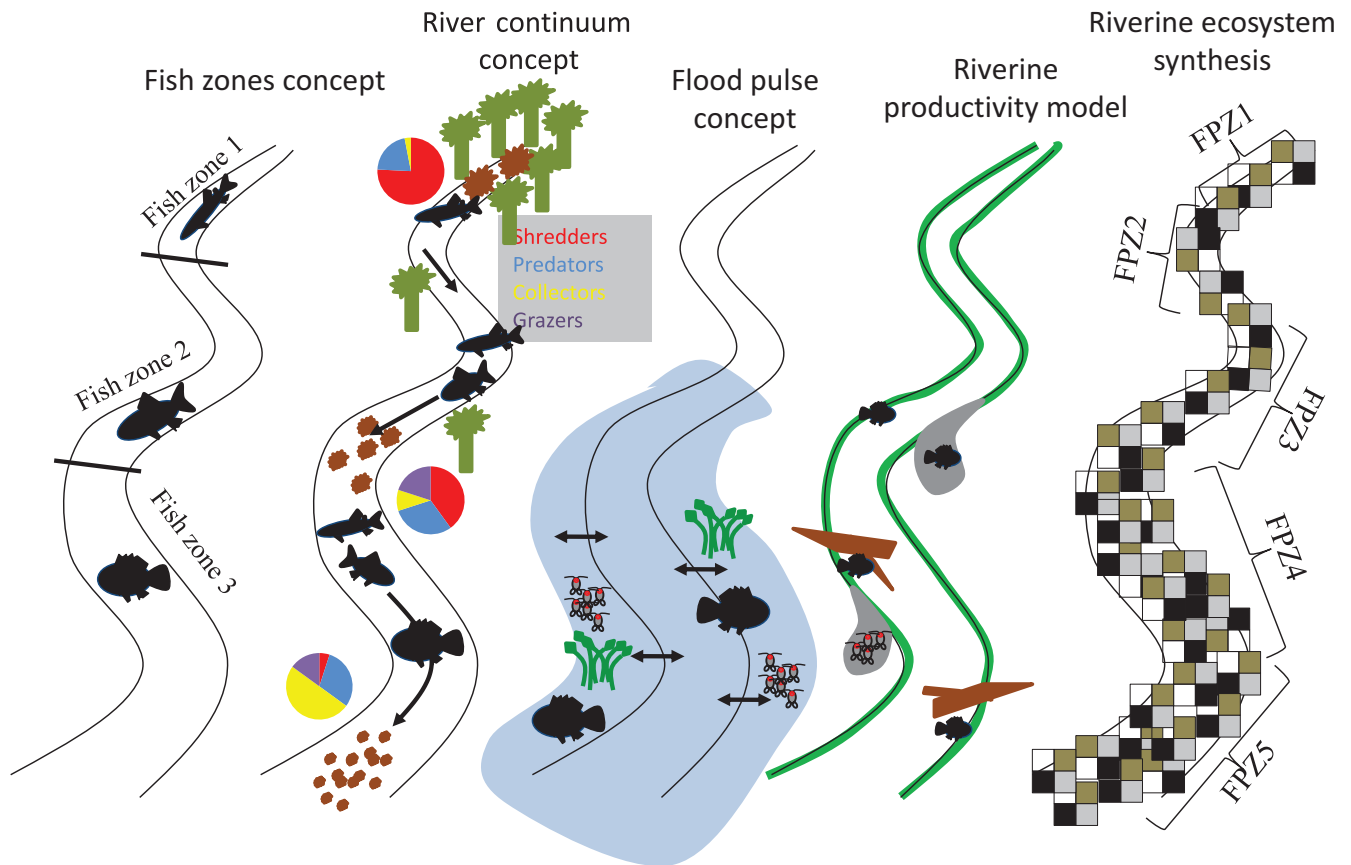


Figure 1. Schematic representations of the main river ecosystem concepts. Abbreviation: FPZ, functional process zone.

pulse is a batch process rather than a continuous process. The riverine productivity model came about because of the perceived underemphasis of the contribution of the edge of large rivers to production, especially in the middle and lower reaches, and the inability of the river continuum concept and the flood pulse concept to adequately explain the structure of food webs (Thorpe and DeLong 1994, Thorpe et al. 1998). The riverine productivity model proposed that, in some river sections, material and energy are derived mainly through the local production of phytoplankton, benthic algae, and other aquatic plants and are derived directly from the riparian zone through leaves and particulate and dissolved organic carbon. The proponents of the riverine productivity model did not, however, reject the river continuum concept or the flood pulse concept completely, suggesting instead that they might be more or less relevant, depending on the river type or section. Subsequent studies have supported the principles of each of these three concepts under differing environmental conditions, locations, and climatic zones (e.g., Bruns and Minshall 1985, Gawne et al. 2007, Hoeinghaus et al. 2007).

As the concepts of river ecosystem functioning have developed, parallel ideas relating to geomorphology and habitat (e.g., the hierarchical framework for stream habitat classification, Frissell et al. 1986; the process domains concept, Montgomery 1999; functional process zones, Thoms

et al. 2004) have been formulated that help underpin and contextualize patterns and processes associated with riverine biota. These concepts are allied to ideas about patch dynamics in lotic systems and how spatial heterogeneity and temporal variability in rivers affect populations, communities, and ecosystems (Winemiller et al. 2010).

In an important development, Walker and colleagues (1995) recognized the potential for incorporating the flood pulse concept and the river continuum concept in modeling river ecosystem patterns and processes and advocated measuring the relative contributions of physical transport and biological transformation to river metabolism (figure 2). Others have also suggested that the river continuum concept, the flood pulse concept, and the riverine productivity model can, at different times and locations, explain sources of energy in rivers (Dettmers et al. 2001). Bunn and Arthington (2002) proposed four principles that link hydrology and biodiversity in rivers, specifically as they related to altered flows, and the riverscapes concept of Fausch and colleagues (2002), at the same time, explicitly appealed for attempts to reconcile the hierarchical nature of streams with the continuous downstream flow of materials and energy and the upstream and downstream links that clearly occur through fish, bird, and insect movement. A significant recent attempt to synthesize river ecosystem

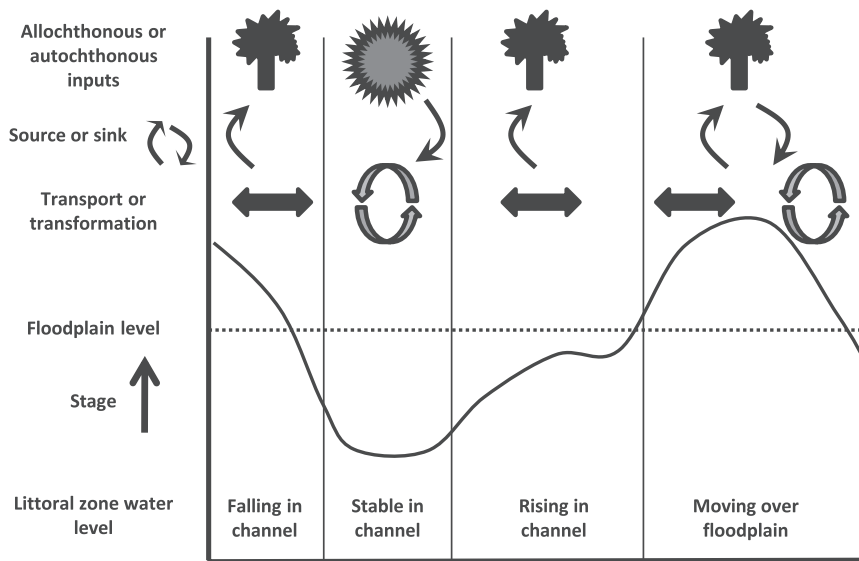


Figure 2. The processes of transport and transformation in relation to phases of the flood pulse for the lower reaches of a turbid floodplain river with a shallow photic zone. Symbols: tree, allochthonous inputs; sun, autochthonous production; upward arrow, source; downward arrow, sink; horizontal double-headed arrow, transport; circular arrows, transformation. Source: Adapted with permission from Walker and colleagues (1995).

concepts was the riverine ecosystem synthesis by Thorp and colleagues (2006, 2008). The riverine ecosystem synthesis was proposed as a merging of ecogeomorphology with a landscape model of hierarchical patch dynamics, but it does not describe the river as a continuum, despite the river ecosystem's being considered in its entirety. Instead, the riverine ecosystem synthesis conceptualizes rivers as "downstream arrays of large hydrogeomorphic patches (e.g., constricted, braided, and floodplain channel areas) formed by catchment geomorphology and climate" (Thorp et al. 2006, p. 1). These patches are equivalent to *functional process zones* (Thoms et al. 2004) and analogous to the *process domains* of Montgomery (1999). The 17 tenets proposed by the riverine ecosystem synthesis relate to the distribution of species and species diversity and the factors that influence these, community (or assemblage) regulation, ecosystem, and riverscape processes (e.g., autochthonous production and allochthonous inputs, nutrient spiraling and life history)—all largely governed by climate, flow and geomorphology. It brings together the river continuum concept, the flood pulse concept, the riverine productivity model, and the various concepts dealing with geomorphic process zones or domains within a nested hierarchical framework.

The strengths of the riverine ecosystem synthesis lie in its ability to bring together river ecosystem concepts, to identify to which parts of the puzzle the concepts belong, and to provide a suite of tenets to progress river science. However, the tenets of the riverine ecosystem synthesis are not testable hypotheses, and the synthesis is complex, which, although it is not an issue in itself, may limit its ease of being understood

and its uptake. The riverine ecosystem synthesis also lacks a general principle underlying flow–geomorphology–ecology relationships that is pertinent to all locations, times, and types of rivers. The authors of the riverine ecosystem synthesis acknowledge that it is a work in progress and encourage improvements and developments. Therefore, there is still room for a concept that synthesizes the existing models in a simple, easily understood, but holistic way; that embeds ecology in hydrology and geomorphology and relates to primary and secondary production and its storage, transformation, and transport through river systems; and that has universal appeal and relevance. We propose the *river wave concept* as such a model.

Our aims in this article are to propose the river wave concept as a means of synthesizing stream ecosystem models into a single, simple concept, based on the properties of a wave; to synthesize stream ecosystem models into the river wave concept and present testable

hypotheses; to describe how river waves interact with their environment; to provide a vehicle for learning for students; and to allow predictions by researchers and managers of natural and altered stream ecosystems.

The river wave concept

We define the river wave concept through the following propositions: The wave provides a useful model for river flow. The location and source of autochthonous production or allochthonous inputs, and the storage, transformation, and transport of the material and energy derived from that production and inputs, are largely a function of the temporal or spatial position (ascending or descending limbs, trough, crest) on the river wave. The nature of river waves is influenced by climate, geology, geomorphology, and anthropogenic regulation and, in turn, influences productivity, biodiversity, and the composition of riverine biota through reciprocal feedback with geomorphological features.

The river wave concept is encapsulated in three hypotheses below. It emphasizes that the key processes that drive river ecosystem structure and function are the production, storage, transformation, and transport of material and energy. The hypotheses emphasize spatial position on the river wave and its significance for river function. Temporal and scale aspects of the river wave concept are also considered below. The river waves that drive ecosystem processes are also responsible for the structure and organization of the physical form of the river and its floodplain. But first, we introduce the conceptualization of river flow as a series of waves.

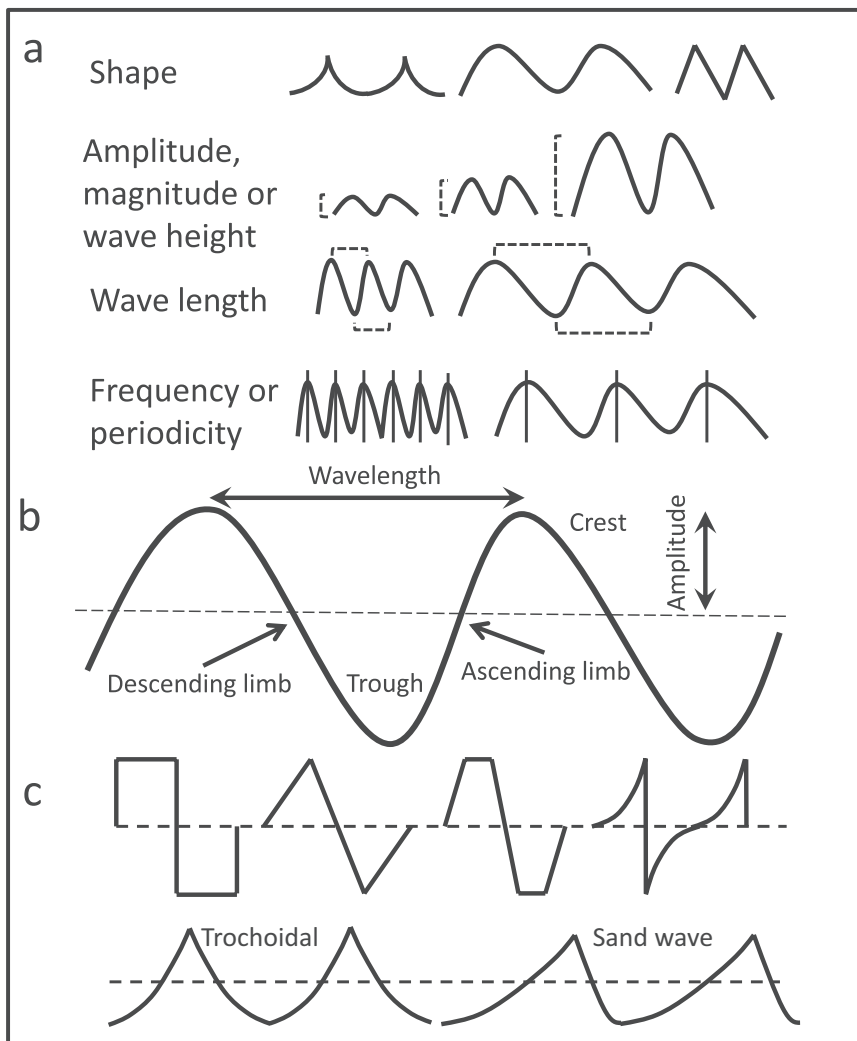


Figure 3. (a) Key characteristics of waves. (b) A sine wave showing the key characteristics of waves. (c) Various waveforms.

River flow can be conceived as a series of waves

Waves are everywhere; they are conceptually simple and have previously been used as models for natural phenomena (Shapiro 1973). Therefore, a wave has potential as a model for river flow, its properties, and its behavior. A wave involves energy (e.g., sound or light) moving through a medium (e.g., water or air), typically without causing the medium to move permanently. A wave can be described by four features: shape, amplitude, wavelength, and frequency (figure 3a). The shape of a wave is a key characteristic: Some waves constitute symmetrical oscillations (figure 3b), whereas others constitute asymmetrical ones (figure 3c). The size of an individual wave can be described by its *amplitude*, or wave height, and its *wavelength*, or the distance between successive crests or troughs (figure 3a). It can also be described in terms of its *frequency* or *period*, the number of wave crests per unit of time or space. The positions on a wave can broadly be delineated as *ascending* or *descending limbs*, *crests*, and *troughs* (figure 3b).

The river wave concept uses the wave as a model, because river flow involves movement of the water itself down an altitudinal gradient. It may be useful to view the river wave as changing river surface elevation through time at a location, keeping in mind that, in the case of the river wave, this occurs as a result of changing volumes of water passing that point with corresponding changes in velocity and *stream power* (its capacity to do work on the physical boundaries), rather than a simple rise and fall. River flow can be conceptualized as a wave at multiple scales (figure 4). At any point in time or space, river flow may be in a trough (*baseflow*), ascending, descending, or at a *crest* (peaking or flooding). As it moves down a river, the river wave's wavelength increases, and its amplitude decreases (otherwise known as *attenuation*).

Hypotheses of the river wave concept

The overarching hypothesis of the river wave concept is that the location and source of autochthonous production and allochthonous inputs, storage, transformation, and the longitudinal or lateral transport of the material and energy derived from that production are largely a function of the position (ascending or descending limbs, trough, crest) on a river wave, either temporally or spatially.

The three secondary hypotheses relate to the patterns and processes associated with troughs, ascending and descending

limbs, and crests of the river wave (figure 5). Therefore, their emphasis is on spatial aspects of the river wave. They are described below and broadly follow the principles of the riverine productivity model, the river continuum concept, and the flood pulse concept, respective of the order in which they are presented here. Indeed, the objective of the river wave concept is to unite the three concepts. Because these concepts are well entrenched in the literature, our justification of them will be brief. Nevertheless, the hypotheses are expressly proposed to be tested empirically.

Hypothesis 1. At the trough of a river wave, local autochthonous production and local allochthonous inputs predominate; the transformation and storage of material and energy are of greater importance than transport, which approximates the predictions of the river productivity model. The trough of a river wave equates to low flow (commonly referred to as *baseflow*) or it could fall to zero flow (figure 5). It is at this time that the river wave concept hypothesizes that

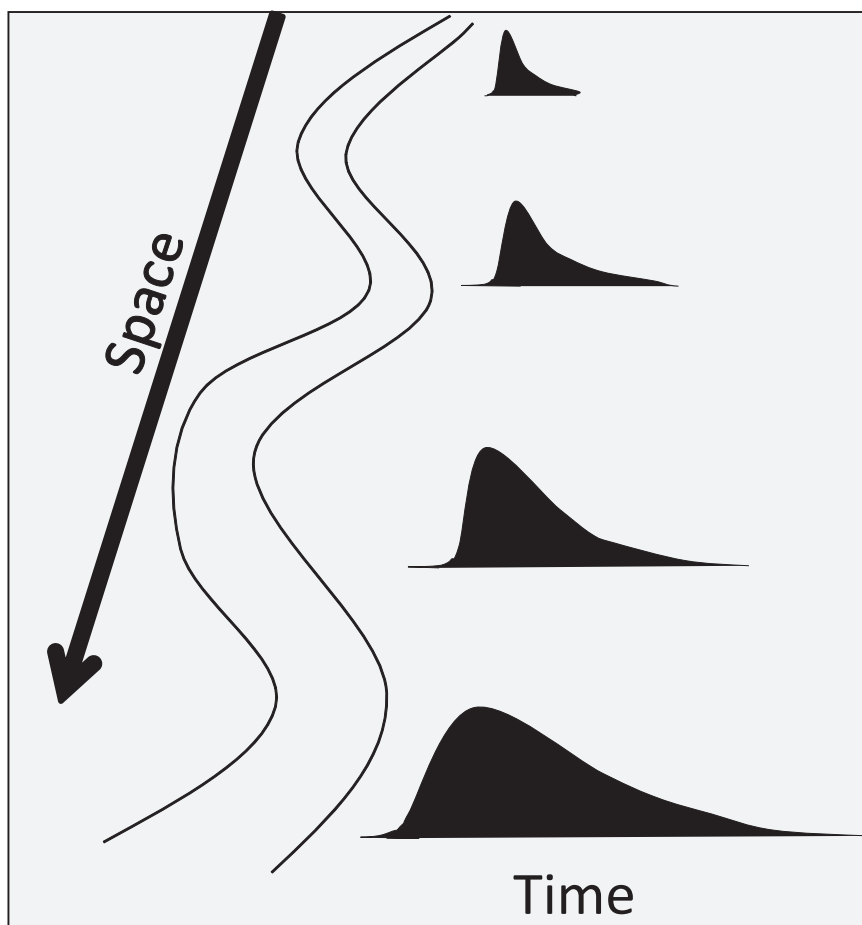


Figure 4. River waves in time and space.

the local production of autochthonous and the local inputs of allochthonous organic matter contribute most to stream metabolism and that significant local transformation of this material through decomposition and assimilation at various trophic levels occurs, which approximates the predictions of the riverine productivity model (Thorpe and Delong 1994, Thorpe et al. 1998).

The rationale for this hypothesis is that when flows are low, the transport of material and energy from upstream is limited, and, in the case of zero flows, transport from upstream is zero, and local production and inputs from riparian sources overwhelmingly predominate (Walker et al. 1995).

Indeed, rivers with intermittent flow are common worldwide (Datry et al. 2014), but the conditions of flow cessation are highly variable, including completely dry systems, rivers that have no visible surface flow but that still carry flow through their bed substrate, and rivers in which only a part of the system may cease to flow. In many cases, these intermittent streams consist of a series of isolated pools along the channel system. Primary production and local allochthonous inputs have been found to be high and, in some cases, greatest during low flows in low- and high-order streams, including the Cinaruco River, in Venezuela (Roelke et al. 2006), and other tropical rivers (Vegas-Vilarúbbia and Herrera 1993);

the River Elbe, in Germany (Wilczek et al. 2005); the River Danube, in Austria (Hein et al. 2003); the Murray River, in Australia (Gawne et al. 2007); and in desert (Bunn et al. 2006) and wet-dry tropic streams generally. It is under these conditions that the degradation of organic material occurs rapidly through microbial and invertebrate activity and carbon transfers up trophic levels (Cotner et al. 2006, Roelke et al. 2006, Gawne et al. 2007). Hoeinghaus and colleagues (2007) concluded that the riverine productivity model best predicted the sources of food web carbon found in high-gradient rivers, rivers downstream of reservoirs, and reservoirs in Brazil during the late dry season, before rivers had started to rise.

Hypothesis 2. On the ascending or descending limbs of river waves, upstream allochthonous inputs and longitudinal transport of material and energy predominate, whereas local production, inputs, storage, and transformation are of lesser importance, which approximates the predictions of the river continuum concept.

The ascending and descending limbs of river waves equate to rising and falling hydrographs, respectively. It is at these times that the river wave concept hypothesizes that rivers are dominated

by upstream allochthonous production and longitudinal transport of material and energy, that storage and transformation of material are of lesser importance, and that the river continuum concept (Vannote et al. 1980) is the most appropriate of the existing models (figure 5).

The rationale for this hypothesis is that when flows are rising (e.g., through snowmelt or rain events), progressively more of the previously dry river channel becomes inundated with particulate organic matter (POM) and dissolved organic carbon (DOC), dissolved organic matter (DOM) and inorganic matter are entrained, and much of this is transported downstream (Raymond and Saiers 2010). Concentrations of DOC, DOM, and POM are generally positively correlated with discharge. Frequently, however, concentrations are higher during rising than falling hydrographs, which reflects the exhaustion of the available supply of sources of these elements, but this varies with the cause of the rise—storm, winter rains, or snowmelt—and with antecedent conditions (Wilson et al. 2013). In some cases, the particulate peaks lag behind the flow peak because of source area distributions in the catchment (Wilson et al. 2013).

In addition to nonliving material, rises and falls in the hydrograph commonly result in increased transport of riverine phytoplankton (Townsend et al. 2012) and zooplankton

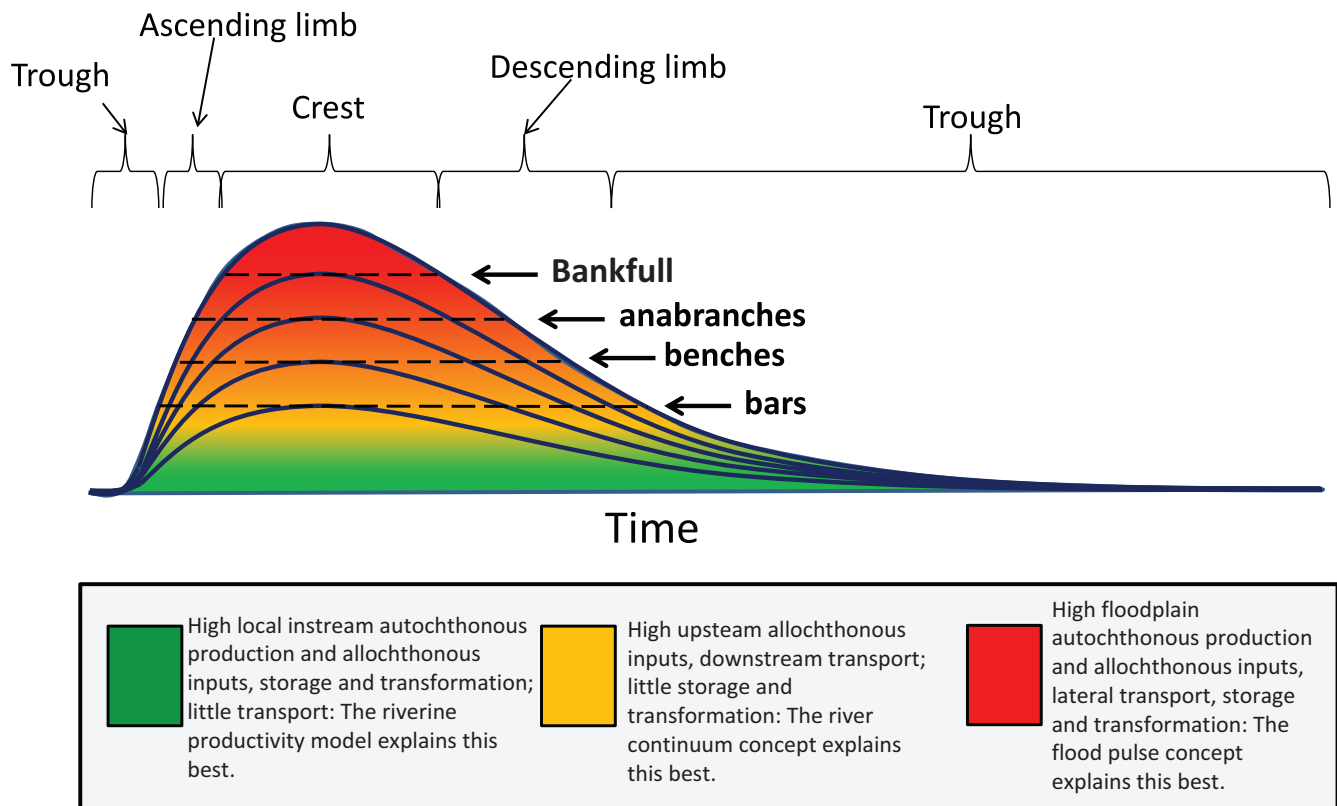


Figure 5. The relative importance of sources of autochthonous production and allochthonous inputs, storage, transformation, and transport of material and energy at the trough, ascending and descending limbs, and crests of river waves.

(Humphries et al. 2013), increased macroinvertebrate drift rates (Brittain and Eikeland 1988), and induced movement and migrations of fish (Lucas and Baras 2001). Lags in the responses of living and nonliving material to ascending and descending limbs of river waves will undoubtedly occur.

Hypothesis 3. As the river wave rises to a crest, the contribution of allochthonous inputs of material and energy from floodplain habitats by lateral transport and then by autochthonous floodplain production increases, storage and transformation of material become of great importance, although upstream allochthonous production and transport continue to be substantial, which approximates the predictions of the flood pulse concept.

The crests of river waves equate to flood flows in rivers, which, for floods approaching and exceeding bankfull, progressively inundate the floodplain proper. It is at these times that the river wave concept hypothesizes that the contribution of allochthonous inputs of material and energy from floodplain habitats, by lateral transport and then by autochthonous floodplain production, dominates in rivers; that the storage and transformation of material play important roles; and that the flood pulse concept is the most appropriate of the existing models (figure 5).

The rationale for this hypothesis is that, as river waves rise to crests, the newly inundated substrate contributes

organic matter to the river and that, at their maximum, when wave crests pass overbank, floodplain inputs dominate. The significance of the flood pulse to tropical, subtropical, and temperate lowland river production has been investigated extensively (Gawne et al. 2007, Hoeinghaus et al. 2007). For low-gradient rivers, it appears that the flood pulse concept can predict the source of carbon in food webs (Hoeinghaus et al. 2007). The dominant source of carbon on the Ohio River floodplain during flooding is decomposing terrestrial plants (Thorp et al. 1998), and DOC input from only 34 square kilometers of the floodplain of the River Murray, in southeastern Australia, during flooding, equated to the amount derived from in-channel algal production in 1 year (Gawne et al. 2007). In the case of the Brazos River, in Texas, however, most of the carbon was derived from riparian macrophytes (Zeug and Winemiller 2008a), which is not inconsistent with the flood pulse concept but which does contradict other studies that emphasize the importance of main-channel algae as the primary source of carbon for large rivers during floods (Delong et al. 2001). It is apparent that flood dynamics (i.e., shape, amplitude, wavelength, and frequency) are the key to the significance of the flood pulse for the storage or movement of organic and inorganic matter between the floodplain and the main channel (Tockner et al. 1999, Peduzzi et al. 2008), primary production (Thorp et al. 1998), transformation of the matter through microbial activity and transference

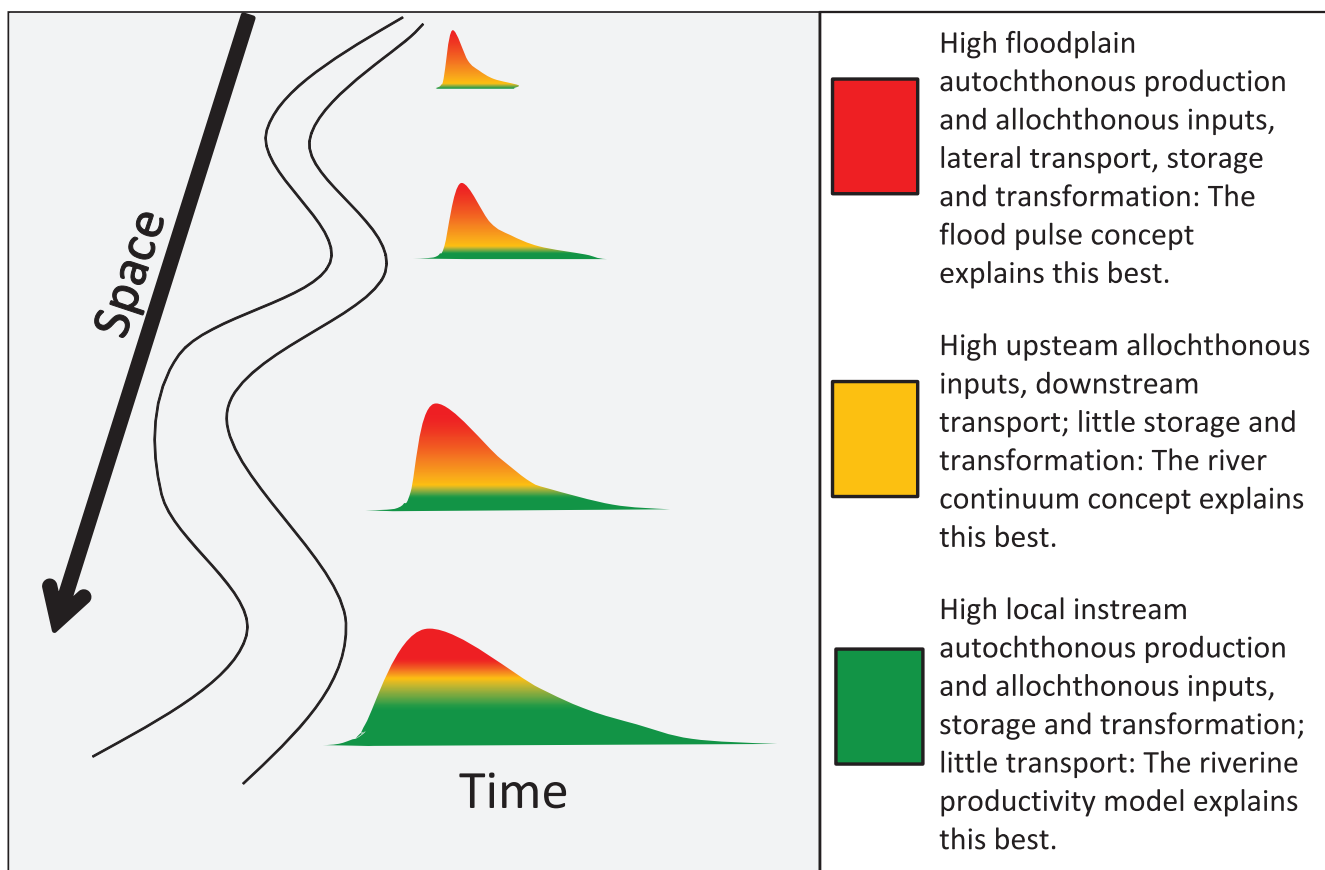


Figure 6. Hypothetical examples of the variation in time and space of river waves and the relative importance of troughs, ascending and descending limbs, and crests to autochthonous production, allochthonous inputs, transport, storage, and transformation of material and energy.

up trophic levels (Gutreuter et al. 1999, Benke et al. 2000, Lindholm et al. 2007), fish movement (Zeug and Winemiller 2008b), and fish recruitment (Tonkin et al. 2008). The way flood dynamics are characterized, however, has much to do with scale (Zeug and Winemiller 2008a): Floods described from daily, weekly, or monthly time steps will give very different shapes, depending on the size of the catchment. Of course, during floods, rivers continue to transport the majority of their organic and inorganic matter downstream within the channel, and, of course, sediment and organic matter will be eroded from the bed, banks, and bars within the channel and deposited on in-channel bars and benches, and significant amounts are deposited on the floodplain.

River geomorphology, the shape of the river wave—particularly, the duration of flood waves—and the relative significance of troughs, limbs, and crests to ecological processes as described in the hypotheses above, will undoubtedly be influenced by the location in the catchment: upland, middle, and lowland reaches (figure 6). Furthermore, the ecological processes, described above, may differ in their reaction times to waves and in their positions on waves (i.e., crests, limbs, or troughs). Leaching of nutrients, transport of organic matter, and movement of fish may take place as soon as river waves change from troughs to ascending

limbs, whereas the hatching of zooplankton and the growth of fish larvae feeding on floodplains may lag days or weeks behind the wave crest. These different reaction times, lags, and hysteresis are built into the three main river ecosystem models. For example, the river continuum concept, by its very nature, considers the downstream transformation of organic matter after prior upstream input or transformation, which must, therefore, involve some time lag as the material moves downstream (Vannote et al. 1980). The flood pulse concept includes the potential use of floodplain-derived nutrients or zooplankton by in-channel biota, which, again, must involve a time lag (Gutreuter et al. 1999). Any use of the river wave concept must therefore account for the potential—indeed, likelihood—of delays in ecological responses to river waves.

Interaction of river waves with their environment

The river waves are directly analogous to the river hydrograph. Most of the hydrological terms are simply translatable to features of a wave. Here, we briefly describe the interaction of river waves with their environment: how they are influenced by climate, geology, geomorphology, and human activity and, in turn, influence the riverscape and its biota (figure 7). Here, the emphasis is on temporal and scale

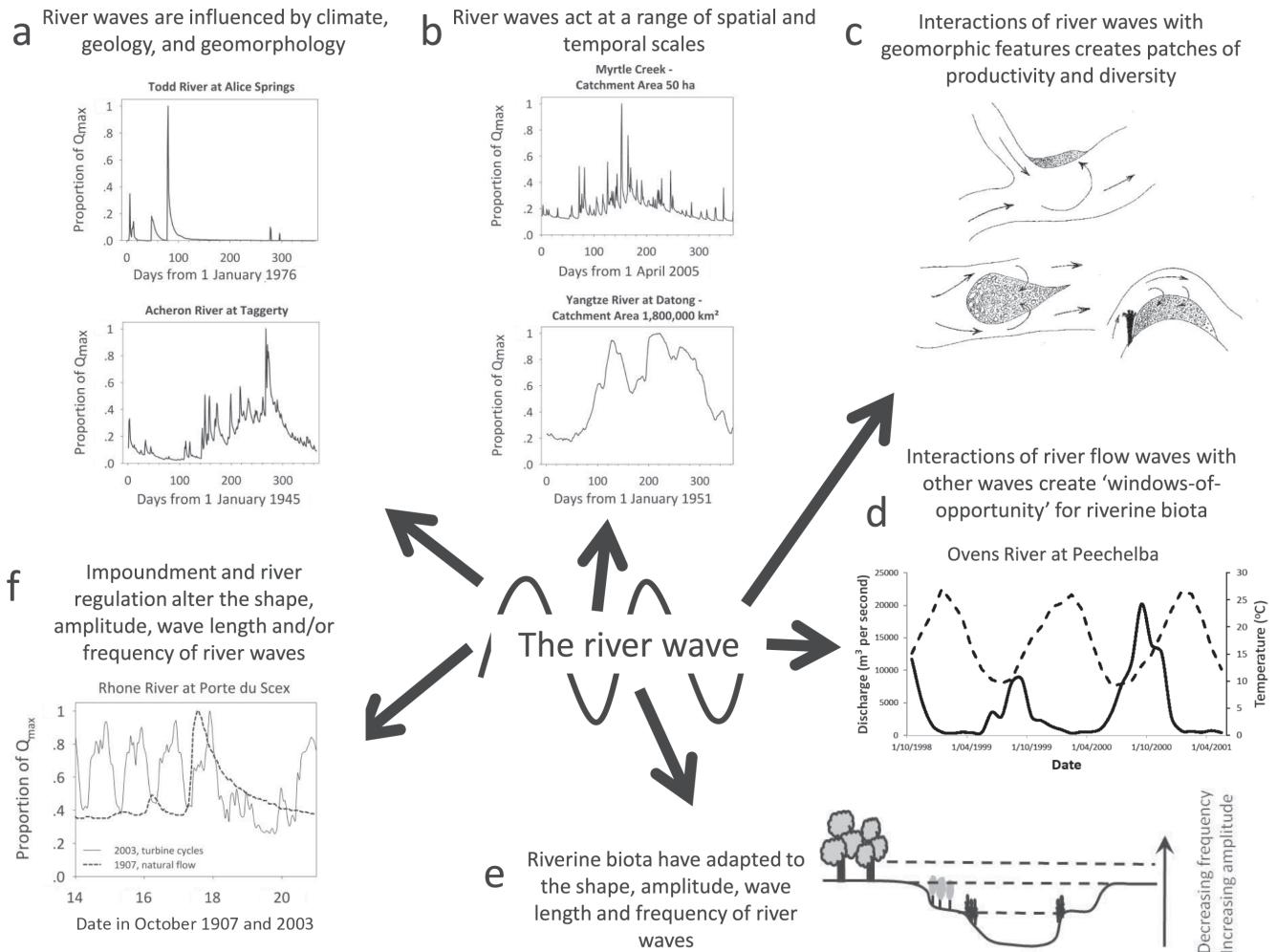


Figure 7. Interaction of the river wave with its environment: (a) Daily discharge as a proportion of the maximum discharge (Q_{max}) for the arid-zone Todd River, at Alice Springs, Central Australia (coefficient of variation [CV] = 1.5) and for the cool-temperate Acheron River, at Taggerty, Southeast Australia (CV = 0.4). (b) Daily discharge as a proportion of maximum discharge for Myrtle Creek, Southeast Australia and the Yangtze River, China. (c) Interactions of river waves with geomorphic features (e.g., confluences of tributaries with the main stem and longitudinal and bank-attached bars). Source: Adapted with permission from Brierley and Fryirs (2005). (d) Interactions of river flow waves with other waves. The solid line represents discharge, and the dashed line represents temperature, in the Ovens River, Peechelba, Southeast Australia. (e) Riverine biota and the shape, amplitude, wavelength, and frequency of river waves; for example, the frequency and amplitude of river waves will influence the types of plants that occur in river channels and on floodplains. (f) Impoundment and river regulation; for example hydroelectricity generation in the River Rhone, Porte du Scex, France, (daily discharge as a proportion of Q_{max}) has altered virtually all features of the river wave (2003, the solid line) from before regulation (1907, the dashed line). Abbreviations: ha, hectares; km², square kilometers; m³, cubic meters; °C, degrees Celsius.

aspects of the river wave. This will reinforce the applicability of the conception of river flow as a wave.

River waves are influenced by climate, geology, and geomorphology.

River waves consist of two fundamental types: flood peaks that arise directly from short-term events in the catchment (e.g., precipitation events, snowmelt) and propagate rapidly through the system, and baseflow waves of long wavelength that reflect the amount of saturated storage in the catchment, usually as groundwater (figure 7a; McDonnell 2003). At the

macro scale, river waves can be conceptualized in terms of the *river regime*, which is defined as the pattern of flows through the year and which is usually assessed on the basis of monthly flows. Natural regimes are largely determined by climate, principally by the seasonal distribution of precipitation, but in many instances, this is substantially modified by temperature. Storage capacity in catchments can also affect the regime relative to climate forcing. There is some variety in the year-to-year consistency of these regimes, especially as regards the magnitude or amplitude of the wave, which

is due to the interannual variability of the flows. The consistency of this pattern declines as the year-to-year variability increases to the point at which, for arid-zone rivers such as the Todd River, in central Australia, with high annual coefficients of variation, the frequency and magnitude of waves are unpredictable, and some years may lack flow altogether.

Impermeable bedrock stores and releases little water, so the baseflow component is small and short-term flow wave responses to precipitation are dominant, whereas highly permeable bedrock absorbs and stores water from precipitation and releases it slowly as baseflow. Steep catchments are more effective in generating rapid flood responses than are low-gradient ones, and the shape of the drainage basin helps determine the time of concentration of flow and, therefore, the speed and intensity of the response to precipitation.

River waves act at a range of scales. The amplitude (the peak flow level) and wavelength (the duration of wave) of river waves are positively correlated with catchment area (figure 7b). The temporal scale of river waves is largely determined by catchment area, although the nature of particular precipitation events is also a driver. In the small streams of the upper catchment, a flood wave may pass in hours, whereas at the lower end of a big river, it may take months. This is a continuum, and the ecological effects will also change along that continuum. The magnitude of the baseflow or trough of waves is also determined by catchment area and geology, whereas the temporal pattern is largely climatically determined. In small catchments, exhaustion of the saturated storage will lead to the cessation of flow more readily than in large catchments. Generally speaking, in unaltered rivers, a cessation of flow begins in the headwaters and then progresses downstream.

Interactions of river waves with geomorphic features creates patches of productivity and diversity. The size and pattern of the river channel and the bedforms that occur within the channel, as well as the nature of the floodplain, are shaped by the passage of waves (figure 7c). Flood waves commonly scour during the rising limb and redeposit during the falling limb of the wave. In fluvial geomorphology, *bankfull discharge* refers to a flow level at which the discharge (or a discharge range) that is considered to be responsible for channel formation and determines the overall channel size. The frequency at which bankfull discharge occurs is typically 1–2 years but can be longer. Floodplains contain a variety of landform features that are a product of the processes by which the floodplain has been built, the age of the individual features, and the way that they have been modified by subsequent overbank flows. There are two basic processes at work in the construction of floodplains: channel migration (expressed as alternating curved swales and depressions and cutoff meanders) and channel avulsion (expressed as abandoned channels) (Nanson and Croke 1992).

Interactions between river waves and riverine geomorphic features are largely encapsulated in the concepts of *process*

domains (Montgomery 1999), *physical biotopes* (Newson and Newson 2000), and *functional process zones*; are central to Bunn and Arthington's (2002) first principle, which relates flow, physical habitat, and biota; and have been termed *large hydrogeomorphic patches* (Thorpe et al. 2006). These include particular channel types, the confluence of tributaries and the main channel, islands, extensive slackwaters and floodplains, and a diversity of in-channel geomorphic features (figure 7c). These patches typically support unique assemblages of plants and animals, and the habitat heterogeneity that results contributes greatly to riverine diversity (Gray and Harding 2007). For example, confluences of tributaries and the main channel (i.e., where two river waves meet) are sites of high diversity and productivity (Kiffney et al. 2006); *slackwaters* (i.e., where river waves are dampened by structure) support abundant microfauna and function as nurseries for shrimp and fishes (Price and Humphries 2010); and *reverse-flow eddies* (i.e., where river waves are reflected) along channel margins can lead to sediment deposition and building of in-channel benches, which may be incorporated into the floodplain (Vietz et al. 2012) and which are important retention zones for instream biota (Schiemer et al. 2001).

Interactions of river flow waves with other waves create windows of opportunity for riverine biota. Flow waves are, of course, only one—but probably in most cases, the dominant—type of wave or pulse that plays a major role in river ecosystem patterns and processes (figure 7d). Thermal waves, for example, may be largely responsible for primary and secondary production and fish breeding in temperate rivers (Humphries et al. 2002). Seasonal resource pulses, such as the occurrence of postspawning salmon carcasses (Helfield and Naimann 2001) or litter fall (Benfield 1997), can drive productivity and influence food webs. Flow waves overlap with these other waves (Valett et al. 2008), creating periods when conditions are most favorable for ecosystem processes and biota, and the life histories of biota have evolved to deal with or take advantage of the relative variability and predictability of these opportunities (Winemiller and Rose 1992, Humphries et al. 2013). In some cases, as in tropical systems, which experience small annual temperature ranges, the influence of flow waves dominates ecosystem patterns and processes (Goulding 1980), whereas in other cases, the roles of flow and temperature are likely synergistic (Tonkin et al. 2011).

Riverine biota have adapted to the shape, amplitude, wavelength, and frequency of river waves. Stream biota have evolved under the dominant influence of the flow characteristics of rivers (figure 7e; Poff et al. 1997, Bunn and Arthington 2002). Flow waves act as strong evolutionary forces influencing life history strategies for all riverine biota, including plants, macroinvertebrates, and fish (Poff et al. 1997, Lytle and Poff 2004). The *shape* of a wave relates to the steepness of its rise and fall, the sharpness of its crest, and its duration. Headwater and temporary streams tend to experience short-duration waves,

with rapid rises and falls and sharp peaks, whereas large rivers tend to experience longer wavelengths, with much slower rises and even more protracted falls, with rounded peaks (figure 7b). Other lotic systems lie on a continuum between these two extremes. Species that inhabit each stream type have adapted through life history and through behavioral and morphological traits to deal with these patterns (Lytle and Poff 2004). Behavioral adaptations allow organisms to avoid rapid flow changes by responding early to heavy rainfall or to increases in current speeds by sheltering and reemerging shortly after or by exploiting them as cues for migration or reproduction (McMullen and Lytle 2012). Morphological adaptations include body shape, which may allow species such as mites, mollusks, and flatworms to shelter from rapid rises in flow when these are associated with intermediate-sized substrates (McMullen and Lytle 2012). Morphological adaptations may also include brittle structures in plants, which break off and so reduce drag in fast currents (Beismann et al. 2000). Long-duration waves with slow rates of rise, characteristic of lowland rivers, create conditions conducive to the use of habitats, such as floodplains, that these waves inundate. Floodplain breeding by tropical species is an obvious example (Goulding 1980, Junk et al. 1989).

Magnitude relates to the *amplitude*, or the highs and lows of flow. Although the heights of floods will vary seasonally and annually, the frequency of particular flood heights can be described for individual river systems and, if they are predictable, will select for life history, behavioral, and morphological traits in riverine biota (Lytle and Poff 2004). For example, in large floodplain rivers, such as the Amazon or the Yangtze, floods of similar magnitude occur every year, and fish take advantage and move out into floodplain lakes and forests to breed and feed (Goulding 1980). In the arid-zone rivers of central Australia, flooding cues fish, such as spangled perch, to leave formerly isolated water holes and to breed on the highly productive floodplain, where young fish feast on zooplankton and where birds, in turn, feast on young fish (Balcombe and Arthington 2009).

Frequency means the return period or the recurrence interval and, to a certain degree, the predictability of particular flow events, such as high and low flows. The degree of predictability of floods and droughts can influence the evolution of stream biota (Lytle and Poff 2004). When extremes, such as low flows or floods, are large, frequent, and predictable, the timing of life histories may evolve such that an organism either avoids or exploits the event (Winemiller and Rose 1992). Use of the floodplain for breeding by fishes during monsoonal flooding is an example of adaptations to predictable highs (Goulding 1980), and fish breeding prior to, during, or immediately after predictable, seasonal drought in temperate dryland rivers is an example of adaptations to predictable lows (Humphries et al. 1999). If the extremes are large and frequent but not predictable (e.g., flooding in southeastern Australia; King et al. 2003), the timing of life histories is unlikely to coincide with these, and bet-hedging

strategies, such as protracted breeding (Winemiller and Rose 1992, Humphries et al. 2013), will probably evolve (Lytle and Poff 2004).

Impoundment and river regulation alter the shape, amplitude, wavelength, or frequency of river waves. River regulation largely takes place for navigation, flood protection, and the operation of instream impoundments—holding and then releasing water when it is needed. The dominant reasons for this operation are navigation (i.e., to allow river traffic to pass upstream and downstream of steep gradients or natural barriers and to maintain sufficient depth throughout the year to allow navigation), hydroelectricity generation (i.e., providing a head of water then released to drive turbines), irrigation supply (i.e., storing water for release into irrigation systems when needed), and domestic water supply (i.e., holding a store of water that is then pumped from the impoundment to where it is consumed).

The way in which the natural river wave will be altered depends on the reason for regulation (figure 7f). Therefore, regulation for navigation will usually do little to the features of the river wave (although smaller ship-induced waves can be harmful to riverbanks and biota) but will, of course, change the way that the wave interacts with the riverscape, and it creates a permanent body of water where one may not have existed before. Where hydroelectricity is used for baseload power, the dam is usually operated as transparent storage in which the inflow is the same as the outflow. The Three Gorges Dam, on the Yangtze River, is a well-known example of this type. Where hydroelectricity is used for peak loads only, the natural wave pattern is mainly eliminated and replaced by a daily cycle of steep rises in flow at the onset of peak-generation time and, similarly, a steep reduction in flow at the end of that period (figure 7f).

The effects on the river wave of regulation for irrigation also varies with the demand or with the downstream supply, but, at its most extreme, it can reverse the seasonality of natural flow regimes and change most features of river waves (Walker 1985). Regulation for irrigation typically reduces the amplitude of flood flows, which are captured in impoundments, and creates truncated peaks when they are released.

Finally, river regulation for domestic water supply can vary from simply providing head to allow pumping where demand exists and no alteration to features of the river wave is necessary (although, as with navigation impoundments, this may change the way that the wave interacts with the riverscape) to a complete cessation of flow. A significant feature of river regulation is that it tends to reduce the severity of low flows and removes significant low-flow events from the flow pattern (McMahon and Finlayson 2003). The serial discontinuity concept of Ward and Stanford (1995) and the principles devised by Bunn and Arthington (2002) describe the effects of dams and flow alteration on patterns of biota and biotic diversity and biotic and abiotic processes in much greater detail.

Learning, researching, and managing rivers using the river wave concept

The river wave concept uses the wave as a model for river flow, because the features of a wave are familiar and easily described and understood, and the concept therefore has great utility for describing river flow. The features of a wave—its shape, amplitude, wavelength, and frequency—equate well to well-established hydrological quantitative descriptors such as flow *shape*, *magnitude*, *return period*, and *frequency*. Furthermore, the positions on a wave—the crest, trough, and ascending or descending limbs—equate to positions on a hydrograph: a flood's peak, baseflow or zero flow, and rising or falling flows. Therefore, it is appropriate to use the concept of a wave to illustrate the salient features of river hydrology.

Because of its simplicity, conceptualizing river flow as a wave can be easily taught in undergraduate-level courses in river ecology, conservation, and management. Using the wave as a model provides the basis for a simple description of river flow and facilitates closer integration of in-stream ecology and stream hydrology. This increases the potential for understanding. Analogies, metaphors, and allegories are well-established and powerful techniques in effective pedagogy (Duit 1991). Indeed, the wave theory of light itself originated from an analogy with water waves (Shapiro 1973).

Our exposition of the overarching hypothesis of the river wave concept—that the location and source of production or inputs and the storage, transformation, and longitudinal or lateral transport of the material and energy derived from that production are largely a function of the position on the river wave—effectively unites the three main river ecosystem concepts: the river continuum concept, the flood pulse concept, and the riverine productivity model. Through this hypothesis, we propose that the three river ecosystem concepts, together, complementarily explain the source of organic matter and the overall nature of storage, transformation, and transport of material and energy in rivers. The secondary hypotheses entail that each concept is more appropriate at different positions on the river wave—the trough, ascending or descending limbs, and the crest. It is beyond the scope of this preliminary presentation to construct a mathematical model of the river wave concept. Others have already produced quantitative procedures, effectively manipulating flow waves for environmental flow allocations (e.g., the Flow Health hydrology assessment tool; <http://io.aibs.org/flowhealth>), so creating a numerical model from the river wave concept is entirely feasible. Nevertheless, for river researchers and managers, the river wave concept allows predictions to be made and hypotheses to be tested, relating to the sources, storage, transformation, and movement of material and energy in rivers at different positions on the hydrograph; for rivers whose waves differ in their shape, amplitude, wavelength, or frequency; and for rivers with natural or altered flow regimes. For example, because dryland rivers are dominated by the troughs of river waves for much of their time, we could hypothesize that local

sources of autochthonous production and allochthonous inputs would predominate and that there would be considerable storage and transformation and little or no transport of this material and energy. Or if the frequency of flood crests of similar amplitudes of one river is lower than those of another river, we could hypothesize that the contribution of allochthonous inputs and autochthonous production from the floodplain would be greater in the former than in the latter and that these differences in river wave characteristics would be reflected in differences in taxonomic and food web structure. Or if a river has its flow altered such that the troughs of river waves are less frequent or of smaller amplitude, we could hypothesize that there would be a greater contribution of upstream sources of organic matter and energy than in an unaltered system. There is also great potential to use the key features of the natural river wave for the restoration of altered flow regimes.

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